Documentation: NL Day-Ahead Energy Price Forecasting Pipeline

1. Data Inputs

The system is fed exclusively with **Known Future Covariates** (features whose forecasts are known in advance for the prediction horizon) at a **15-minute resolution**:

- **Netherlands (NL):**
  - Total demand load forecast (`load_forecast`)
  - Solar generation forecast (`solar_fc`)
  - Onshore wind power forecast (`wind_on_fc`)
  - Offshore wind power forecast (`wind_off_fc`)
- **Germany (DE_LU - Cross-Border Signals):**
  - German solar and wind generation forecasts (`solar_fc_de`, `wind_on_fc_de`, `wind_off_fc_de`)
  - German Day-Ahead market price (`price_de_lu`, resampled from hourly to 15-minute resolution via forward-fill)
- **Temporal Features:** Hour, minute, day of week, month, and a weekend indicator (`is_weekend`)

2. Data Sources

- **ENTSO-E Transparency Platform API:** All data is fetched automatically via the `entsoe-py` library by calling official endpoints (e.g., `query_load_forecast`, `query_wind_and_solar_forecast`, `query_day_ahead_prices`)
- **Local Database / File:** A central `CSV` file (`nl_entsoe_day_ahead_15min_oct2025_to_now.csv`) serving as the historical database for model training

3. Data Synchronization Frequency

- **Historical / Cold Start:** Updated incrementally or in bulk from the platform when a refresh of the training dataset is required
- **Real-time Inference:** During each pipeline execution, the latest available market forecasts are retrieved

4. Target Variable

- The Dutch Day-Ahead electricity price (NL Day-Ahead Electricity Price in €/MWh), fetched via the `query_day_ahead_prices` method for the `NL` bidding zone

5. Forecast Horizon

- Generates predictions for a **2-day horizon (48 hours)** in **15-minute steps** (totaling 133 timestamp points per execution)

6. Execution Frequency

- Designed to run automatically (e.g., via Docker container or a scheduler) daily, right after the Day-Ahead market publishes new data, saving the final output to a results file (`realtime_forecast_2days.csv`)